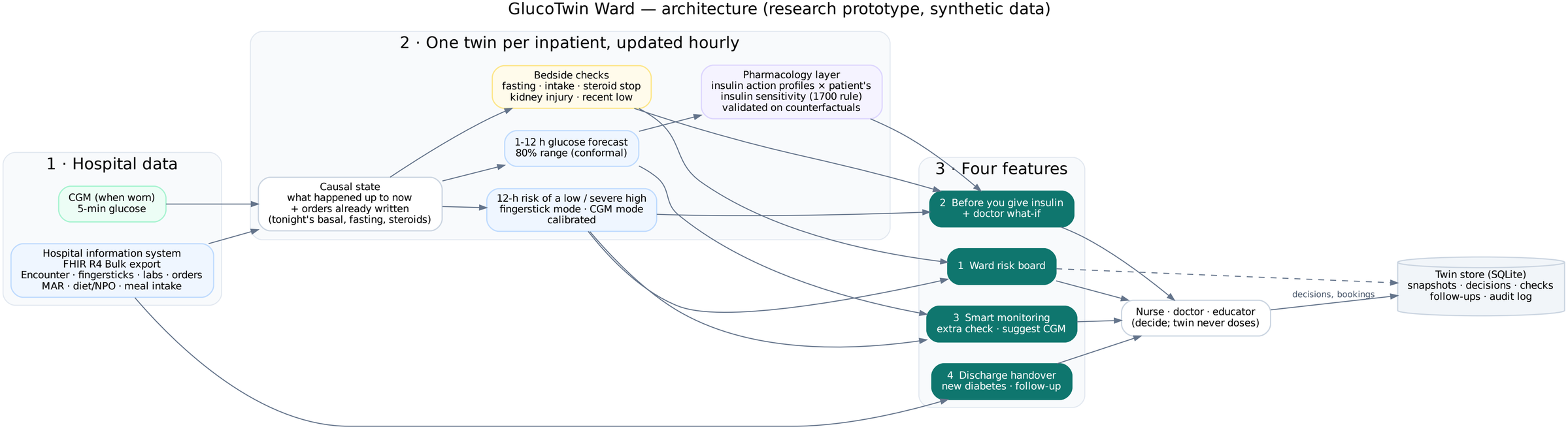
GlucoTwin Ward — architecture (research prototype, synthetic data)
1 · Hospital data
CGM (when worn)
5-min glucose
Hospital information system
FHIR R4 Bulk export
Encounter · fingersticks · labs · orders
MAR · diet/NPO · meal intake
2 · One twin per inpatient, updated hourly
Causal state
what happened up to now
+ orders already written
(tonight's basal, fasting, steroids)
Bedside checks
fasting · intake · steroid stop
kidney injury · recent low
1-12 h glucose forecast
80% range (conformal)
12-h risk of a low / severe high
fingerstick mode · CGM mode
calibrated
Pharmacology layer
insulin action profiles × patient's
insulin sensitivity (1700 rule)
validated on counterfactuals
3 · Four features
2 Before you give insulin
+ doctor what-if
1 Ward risk board
3 Smart monitoring
extra check · suggest CGM
4 Discharge handover
new diabetes · follow-up
Nurse · doctor · educator
(decide; twin never doses)
decisions, bookings
Twin store (SQLite)
snapshots · decisions · checks
follow-ups · audit log